

Clinical Trial Data Management & Quality Dashboard

Self-directed portfolio project using simulated clinical-trial datasets, Excel Power Query and Power BI

1. Project Objective:

To practice a practical clinical data management workflow by reviewing and cleaning clinical-trial datasets, working with SDTM-style clinical domains, performing basic data-quality checks, creating domain-level summaries, identifying missing and abnormal data, and developing an interactive Power BI dashboard for clinical-data review.

2. Domains Used

Domain	Purpose
DM	Demographics
AE	Adverse Events
LB	Laboratory Tests
VS	Vital Signs
CM	Concomitant Medicines
SV	Subject Visits

3. Workflow:

Raw clinical CSV files → Excel / Power Query → data cleaning & type checks → missing/duplicate/QC checks → domain summaries → Power BI dashboard

4. Tools Used:

Microsoft Excel, Power Query, Power BI and GitHub, with SDTM-style clinical-domain concepts and basic data-quality analysis.

5. Key Analysis & Findings:

Demographics (DM) - 306 subjects were analysed. The dashboard reviews age, sex, race, treatment-arm distribution and study duration.

Adverse Events (AE) - 1,193 AE records were reviewed. Severity counts included 770 mild, 378 moderate and 43 severe records. Three records were serious and three had a fatal recorded outcome. Two records had missing seriousness and two had missing outcome.

Laboratory Tests (LB) - Laboratory results were summarized by test and reference-range status. Recorded categories included 56,892 normal, 860 low, 1,505 high, 318 abnormal and 5 blank results. Low + High + Abnormal total 2,683 records.

Vital Signs (VS) - Vital-sign records were summarized by test to review the distribution and coverage of recorded measurements.

6. Power BI Dashboard:

Page 1 - Clinical Trial Data Management Overview Key subject and AE metrics, AE severity/outcome, laboratory result status and vital-sign data coverage.

Page 2 - Subject Demographics Total subjects, age, sex, race, treatment-arm distribution and study duration, with interactive demographic slicers.

Page 3 - Data Quality & Clinical Insights Missing AE seriousness, missing AE outcome, fatal AE records, laboratory findings and vital-sign coverage.

7. Data Quality:

Focus The project practiced identifying missing values, blank categories, data-type inconsistencies, duplicate records, abnormal laboratory findings, domain-level record distributions and basic consistency checks between raw data and summarized outputs.

8. Key Learning Outcomes:

Clinical-domain data handling; SDTM-style dataset structures; Excel Power Query; data cleaning and transformation; basic clinical data QC; summary-table generation; Power BI visualization; interactive filtering; and translating clinical datasets into understandable data-quality insights.

9. CDM Relevance:

The project provides hands-on practice with a simplified clinical data workflow: raw data review, transformation, domain-level analysis, quality checks and reporting. In production CDM environments, this workflow is complemented by validated systems, CRF/EDC design, edit checks, data queries, coding, reconciliation, SDTM mapping, validation and database-lock processes.

10. Project Scope:

This is a self-directed portfolio project using simulated/educational clinical-trial data. It demonstrates clinical data-management concepts and workflow practice and is not a production clinical-trial database or regulatory-submission project.

Author: Sayed Munazza Saniya | M.Sc. Biotechnology | Clinical Data Management and Drug Safety Professional